

Nick Nunez

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SUMMARY

Software engineer with 3 years of professional experience building production web applications and a growing focus on systems programming in C/C++. Currently building real-time audio projects for macOS using CoreAudio, CoreMIDI, Lua embedding, and custom C/C++ libraries, with an emphasis on API design, platform-level learning, and clean host/engine boundaries.

PROJECTS

Standalone Groovebox for macOS

github.com/nico-nunez/meh-groovebox-visceral

- Building a standalone C/C++ performance instrument for macOS with shared transport, pattern-based sequencing, multi-track synthesis, and internal mixing.
- Integrated directly with CoreAudio and CoreMIDI, handling audio device setup, buffer callbacks, MIDI input, and real-time event scheduling without JUCE or RtAudio.
- Embedded Lua as a live control surface for parameter changes, sequencer control, performance scripting, and authored project documents.
- Hosts a reusable synth engine with per-step parameter locks, preset management, and a built-in editor with language server support.

Embeddable C/C++ Synthesizer Library

github.com/nico-nunez/meh-synth-engine

- Built a reusable C/C++ synth engine extracted from an earlier standalone synth app and designed for embedding in multiple hosts.
- Implemented 64-voice polyphony with wavetable oscillators, envelopes, tempo-synced LFOs, FM routing, effects, and a 16-slot modulation matrix.
- Designed the rendering path to separate audio-rate rendering from control-rate updates, decoupling host buffer sizes from internal processing and avoiding unnecessary work in the real-time hot path.
- Exposed an event-driven API for MIDI events, scalar parameter updates, higher-level engine commands, and full-state preset serialization.

EXPERIENCE

React Developer / Software Engineer | Publicis Media

September 2022 - November 2025

- Led frontend development across multiple enterprise applications for internal teams and external clients, growing from sole frontend developer into a technical leadership role.
- Built production React/TypeScript applications for analytics and campaign management workflows, including dashboards handling 10-50K records per session.
- Established shared component patterns, reviewed code, mentored developers, and supported hiring for junior and mid-level frontend roles.
- Collaborated with Java/Spring Boot and Django/Python backend systems by designing API contracts, integrating data pipelines, and diagnosing cross-service issues.

Lawn Care Technician | TruGreen

2013 - 2021

- Advanced from residential to commercial accounts over 8 years before transitioning into software engineering through self-directed study.

TECHNICAL SKILLS

- **Systems:** C/C++ (C++17), real-time programming, static libraries, Make

- **Audio / Platform APIs:** CoreAudio, CoreMIDI, macOS system frameworks
- **Scripting / Embedding:** Lua, shell scripting
- **Web / Professional:** TypeScript, React, Python, Django, Node.js, PostgreSQL
- **Tools & Infrastructure:** Git, Docker, AWS, GCP, Vite, Jenkins

EDUCATION

High School Diploma